THE UNIVERSITY OF SYDNEY

LIFE EXPECTANCY ANALYSIS

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EXECUTIVE SUMMARY

This report investigates global life expectancy patterns using cross-sectional data from 163 countries in 2015. Correlation and regression analysis identify key determinants influencing life expectancy, with adult mortality, income composition of resources, schooling, and HIV/AIDS emerging as the strongest predictors. South Africa is selected for focused policy analysis due to its high adult mortality and health inequality challenges despite its moderate-income composition of resources. Policy recommendations target reducing adult mortality and expanding HIV/AIDS treatment access to foster improvement in national life expectancy.

INTRODUCTION

Life expectancy is a fundamental indicator of national well-being, shaped by healthcare systems, socioeconomic conditions, and public health outcomes. This report analyses global trends in life expectancy (2015), and identifies the key variables most strongly associated with it through statistical correlation and multiple regression modelling. Based on these findings, South Africa is selected for focused country-level analysis to evaluate how structural inequalities and health burdens influence life expectancy outcomes. This report produces targeted, evidence-based policy recommendations and predictive estimate of the potential improvements attainable in South Africa through strategic health interventions.

DATA PREPARATION AND EXPLORATORY ANALYSIS

Firstly, countries containing any missing values in retained variables were removed to ensure consistent comparisons and avoid imputation assumptions. While transparent, this choice may reduce the sample size, which could impact the external validity and generalizability of the analysis, potentially introducing bias to the dataset.

Table 1: Cleaning Data Summary

Item	Value
Countries before cleaning	183
Countries removed	20
Final countries	163

Note: Three erroneous variables were identified, with the first being Alcohol due to extensive missing values. Very high BMI values and Measles values exceeding 1,000 cases suggest inconsistencies.

Exploratory Analysis

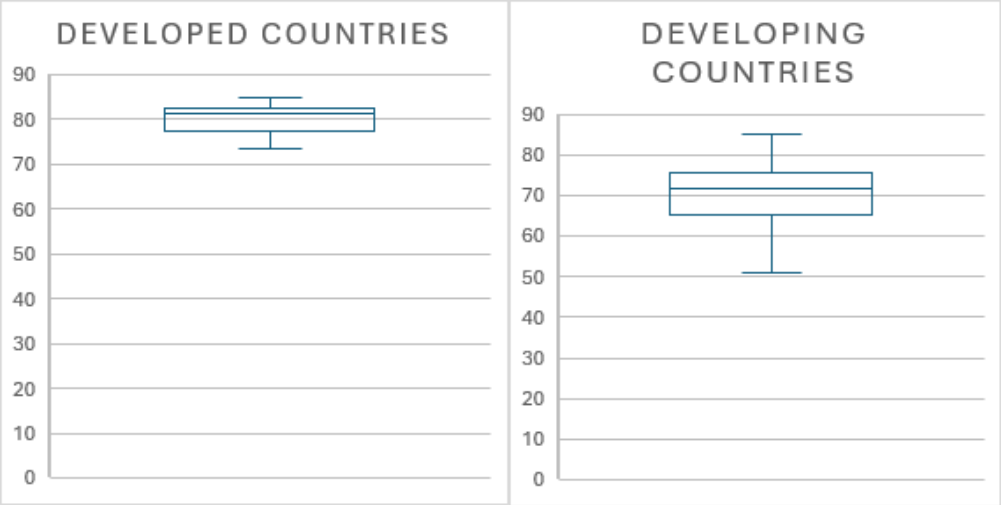
Through an initial analysis of central tendency across all retained variables, we obtained a concise statistical baseline for how health, immunization, and mortality indicators differ across countries in 2015. For each measure, we calculate the central tendency using EXCEL functions, namely =AVERAGE, =MEDIAN, and =MODE.SNGL, and =SKEWNESS, and comment on the distributional shape and dispersion.

Table 2: Descriptive Analysis across all variables

Variable	Mean (4 d.p)	Median (4 d.p)	Mode (4 d.p)	Descriptive Analysis
Life expectancy (years)	71.3031	73.5	75	Countries cluster in the mid-high 70s with a longer lower tail. Mild negative skew (skewness ≈ -0.55).
Adult mortality (deaths per 1000)	153.8405	142	118	Right-skewed, with a minority of high-burden countries pulling the mean above the median.
Infant deaths (deaths per 1000)	23.3620	2	0	Extremely right-tailed, most of the countries are zero .
Alcohol (litres per capita)	3.378	1.79	1.79	Median below mean indicates right-skew (calculated based on countries with data).
Percentage expenditure (% GDP per capita)	2.67641	0	0	Highly sparse with many zeros.
Hepatitis B (% of 1-year-olds immunized)	82.4724	93	99	This data indicates negative skew, concentrated in the sub-90% tail.
Measles (per 1000 population)	1580.0491	16	0	Highly-skewed, some values >1,000 suggest inconsistency.
BMI (average across population)	42.4546	48.6	19.1	Some values appear implausibly high for a population mean, suggesting inconsistency.
Under-five deaths (deaths per 1000 live births)	30.8712	3	0	Heavy right-tailed.
Polio (% of 1-year-olds immunized)	82.5337	93	99	High central tendency at the 90% sub-tail.
Diphtheria (% of 1-year-olds immunized)	84.2147	93	99	Cluster near the top (median 93, Q3 97).
HIV/AIDS (deaths per 1000 live births ages 0-4)	0.6791	0.1	0.1	Very low in most countries, some countries are elevated. Higher values likely reduce life expectancy.
Thinness 1-19 years (% prevalence)	4.7282	3.5	1.9	Right-skewed, mass at low values with a long upper tail, likely negative association with life expectancy.

Thinness 5-9 years (% prevalence)	4.7883	3.5	6.1	Right-skewed, clustering around 6.1% with a small tail, likely negative association with life expectancy.
Income composition of resources (income component 0-1 scale)	0.6842	0.719	0.637	Higher central tendency and tighter spread among developed countries, suggesting a visible social-economic gradient.
Schooling (Average years of schooling)	12.8049	13	12.7	Moderate spread reflecting development gaps, anticipated positive association with life expectancy.

Graph 1: Developing vs Developed countries Life expectancy



The figure illustrates a clear dispersion gap between two groups: developing and developed countries. While developed countries cluster high with a narrow spread (mean ≈ 80.0909 , median ≈ 81.45, standard deviation $\approx 3.31, N = 22$), developing countries sit lower with greater variability (mean ≈ 69.9319, median ≈ 71.8, standard deviation $\approx 7.20, N = 141$). The difference in life expectancy between the two groups is ≈ 10 years, indicating a substantive disparity in the countries' levels. Mortality indicators (adult, infant, under-five) are strongly right-skewed, consistent with the lower tail in life expectancy. At the same time, immunization measures tend to exhibit a ceiling effect, with values clustering in the 90-100% range. Furthermore, the socio-economic measures (income composition, schooling) are higher and tighter in developed settings, aligning with the life expectancy gap. Overall, these patterns point to three matters: mortality rate, immunization, and socio-economic status.

CORRELATION ANALYSIS

In this section, we present a correlation analysis, which serves to identify how the variables relate to life expectancy within our dataset.

Table 3: Correlation Matrix

Variables	Life expectancy	Adult mortality rate	Infant mortality rate	Death under age 5	Income index	Education years
Life expectancy	1,00					
Adult mortality rate	-0,89	1.00				
Infant mortality rate	-0,85	0.82	1.00			
Death under age 5	-0,86	0.80	0.95	1.00		
Income index	0,78	-0.70	-0.72	-0.74	1.00	
Education years	0,75	-0.65	-0.66	-0.68	0.84	1.00

Based on the comprehensive analysis above. We proceed with the following evaluation. There is not just one factor that affects life expectancy. It can be roughly divided into three parts: fundamental causes, external causes, and the consequences. Income determines a country's fundamental economic level, while education determines its development level. Therefore, income is the fundamental factor. Disease and immunisation as external factors. It can be intervened to affect life span. The higher the mortality rates of adults and children in a country, the lower the average life expectancy. This is data that can directly demonstrate the results.

Next, pairwise correlations among potential explanatory variables. National wealth demonstrates a strong positive correlation with educational attainment. Similarly, enhanced educational levels contribute substantively to economic development. These two factors are in an inseparable relationship. Infant mortality and under-five mortality represent overlapping demographic measures, as the infant cohort falls entirely within the under-five age group. Infants constitute a demographic subset within the under-five population. Either of the two viral vaccines can reflect the basic vaccination of a country.

High correlations between explanatory variables will pose problems for regression analysis. The meanings expressed by the two explanatory variables are essentially the same. There is no way to separate them and discuss each one individually. They occurred in the same proportion.

1. Coefficient interpretation: Originally, the coefficient was the same for the increase or decrease of both factors. However, due to their high collinearity, the estimated effects.
2. Statistical significance: Due to its high correlation, it will significantly increase uncertainty.

3. Model stability: By using different models for calculation, when the data is updated appropriately, the models become less accurate.

CONFIDENCE INTERVAL ANALYSIS

We estimate the mean life expectancy across the 163 countries retained after cleaning using a 95% confidence interval. A normal approximation for the sampling distribution of the mean may be reasonable, provided the observations can be treated as independent. This does not mean that the underlying life expectancy data are normally distributed.

$$CI = \bar{x} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

Where

CI = confidence interval
$\bar{x}$ = sample mean
$z_{\frac{\alpha}{2}}$ = confidence level value
s = sample standard deviation
n = sample size

For global average life expectancy:
$\bar{x}$=71.3031
z =1.9602
s =7.6399
n =163

$\therefore$ 70.1301 $\leq \bar{x} \leq$ 72.4761

Interpretation: We are 95% confident that the true global average life expectancy in 2015 lies between 70.13 and 72.48 years (2 d.p). This reflects the combined influence of both developed and developing nations.

Variable Chosen for Hypothesis: Income composition of resources

Income composition of resources reflects a country's capacity to convert economic output into equitable access to essential services such as healthcare, education, nutrition, employment and social infrastructure. Academic research establishes a strong, positive relationship between income and life expectancy. Preston's (1975) analysis found that national income per head remains a primary predictor of mortality reduction, accounting for up to 80% of the variance in average life expectancy globally when adjusted for inequality. Similarly, Peters et al. (2008) finds that higher income composition enables governments to invest in universal health systems and social welfare programs, reducing preventable deaths and strengthening population resilience. The McKinsey Global Institute (2023) provides contemporary reinforcement of these arguments, demonstrating that nations with sustained improvements in income distribution and social investment retain up to 10 years longer average lifespan in contrast to countries with similar GDP, but weaker resource equity. McKinsey asserts that progress in human health increasingly depends on both economic growth, and how income is distributed and reinvested into health, education and infrastructure. Therefore, these findings justify the use of income composition of resources as an explanatory variable for life expectancy, and indicate a positive relationship between the two, with higher values corresponding with longevity.

<u>Hypothesised Relationship</u>

Direction: Positive - as income composition rises, life expectancy should increase.

Magnitude: moderate to strong effect; academic scholarship consistently reveals socioeconomic status to be a primary determinant of population health and longevity.

Subgroup Definitions

High-income composition: ≥ median value
Low-income composition: < median value

Life Expectancy by Income Composition Subgroup Calculations

Median income composition = 0.7190 (4 d.p)

Table 4: Subgroup Confidence Intervals

Group	n	Mean (x)	Standard Deviation (s)	95% CI Lower	95% CI Upper
Low Income Composition	81	65.6358	6.3495	64.2529	67.0187
High Income Composition	82	76.9012	3.6160	76.1185	77.6840

Statistical Interpretation

The 95% confidence intervals for the two subgroups do not overlap:

Low income composition [64.25, 67.02]

High income composition [76.12, 77.68]

The confidence intervals for high-income and low-income composition groups do not overlap, indicating a statistically significant difference in life expectancy. The approximate 11-year gap in mean life expectancy between the two income composition groups reinforces global health inequality. This supports the hypothesis that higher income composition - reflecting stronger economic capacity and equity - is associated with significantly greater life expectancy.

HYPOTHESIS TESTING
We conduct a two-sample hypothesis test comparing the mean life expectancy between countries with High income composition (mean ≥ 0.7190) and Low income composition (mean ≤ 0.7190).

Hypotheses

Let group H be countries with high-income composition and group L be countries with low-income composition.

- $H_0: \mu_H - \mu_L = 0$ (no difference in mean life expectancy)
- $H_\alpha: \mu_H - \mu_L \neq 0$ (means differ)

Test choice

We use a two-sample variance Welch's t-test to compare the mean life expectancy of the abovementioned subgroups since Welch's test does not assume equal variances and offers better

Type I error control than the pooled t-test. However, according to Ahad & Yahaya in 2014, Welch's test is not universally robust when non-normality, variance heterogeneity, and unequal group sizes occur together, especially with negative pairing (when the smaller group has the larger variance), the Type I error can inflate above nominal levels. This occurrence may not have appeared in our data group due to the essentially balanced group size ($n_H = 82, n_L = 81$). In addition, although variances differ, the large, nearly equal sample sizes and CLT (Central Limit Theorem) considerations ($n \geq 30$) make the Welch's test appropriate.

Calculations

- Standard error:
  - $SE = \sqrt{\frac{S_H^2}{n_1} + \frac{S_L^2}{n_2}} = \sqrt{\frac{3.6160^2}{82} + \frac{6.3496^2}{81}} \approx 0.8107$
- Welch t-statistic:

$$t = \frac{\bar{x}_H - \bar{x}_L}{SE} = \frac{76.9012 - 65.6358}{0.8107} \approx 13.8962$$

- Degree of freedom:

$$df = \frac{\left(\frac{S_H^2}{n_H} + \frac{S_L^2}{n_L}\right)^2}{\frac{1}{n_H - 1}\left(\frac{S_H^2}{n_H}\right)^2 + \frac{1}{n_L - 1}\left(\frac{S_L^2}{n_L}\right)^2} = \frac{\left(\frac{3.6160^2}{82} + \frac{6.3496^2}{81}\right)^2}{\frac{1}{82 - 1}\left(\frac{3.6160^2}{82}\right)^2 + \frac{1}{81 - 1}\left(\frac{6.3496^2}{81}\right)^2} \approx 127$$

- P-value:
  - =T.DIST.2T(13.8962, 127) ≈ 0.0000
  -
- 95% CI (confidence interval) for the mean difference:
  - Critical value: =T.INV.2T(0.05, 127) ≈ 1.9788
  - CI:

$$\bar{x}_H - \bar{x}_L \pm t_{\frac{127, 0.05}{2}} * SE = [\,9.6612, 12.8696]$$

Interpretation

At the 5% significance level, we reject the null hypothesis of equal mean life expectancy. The two-sided Welch's t-test provides evidence that mean life expectancy differs between the two groups. The high-income-composition group had a higher observed mean (76.90 versus 65.64 years), with an estimated difference of 11.27 years (95% CI: [9.66, 12.87]). This indicates an association rather than a causal relationship.

REGRESSION MODEL DEVELOPMENT

Model A

Regression Statistics of Model A	
Multiple R	0.943281
R Square	0.889778
Adjusted R Square	0.884053
Standard Error	2.601479

Observations	163

	Coefficients	*P-value*
Intercept	48.84709	1.64E-65
Adult Mortality	-0.01777	8.47E-08
infant deaths	0.045496	0.044198
under-five deaths	-0.03742	0.034388
HIV/AIDS	-0.6399	0.001605
Hepatitis B	0.034915	0.095114
Polio	0.011625	0.296705
Diphtheria	-0.01111	0.64409
Income composition of resources	33.3449	7.64E-38

Approach & Models

Model A includes adult mortality (AM), infant deaths (ID), under-five deaths (UD), HIV, Hepatitis B (HB), Polio (P), Diphtheria (D), and income composition (IC).

$$LE = \beta_0 + \beta_1 AM_i + \beta_2 ID_i + \beta_3 UD_i + \beta_4 HIV_i + \beta_5 HB_i + \beta_6 P_i + \beta_7 D_i + \beta_8 IC_i + \epsilon_i$$

The coefficients of adult mortality, Infant/under-5 deaths, and HIV are negative, while the three vaccine immunization (HepB/Polio/DTP3) and income composition are positive. This model covers three sections: mortality burden, immunization, and socio-economic circumstances. The largest and most reliable effects on life expectancy are adult mortality ($\beta \approx -0.0178, p < 0.0001$), HIV ($\beta \approx -0.64, p \approx 0.016$), and income composition ($\beta \approx 33.34, p < 0.001$). However, two of the groups of predictors are redundant and introduce multicollinearity. The two groups are (i) infant deaths and under-five deaths (infant is a subset of under-five), and (ii) three vaccines (HB, P, D), all indicating immunization performance, and therefore are highly correlated. This redundancy might inflate the coefficient standard errors, weaken individual p-values, can produce the conflicting values of child-mortality signs (infant is positive while under-five is negative).

Model B

Regression Statistics	
Multiple R	0.941254
R Square	0.885959
Adjusted R Square	0.882327
Standard Error	2.620761
Observations	163

	Coefficients	*P-value*
Intercept	48.58208	1.33E-65
Adult Mortality	-0.01775	8.82E-08
under-five deaths	-0.00203	0.287574
HIV/AIDS	-0.68759	0.000698
Income composition of resources	33.22742	2.55E-38
Immunisation	0.039101	0.000673

$$LE = \beta_0 + \beta_1 AM_i + \beta_2 UD_i + \beta_3 Imm_i + \beta_4 HIV_i + \beta_5 IC_i + \epsilon_i$$

$$Immunisation = \frac{HB + P + D}{3}$$

To address this issue, we collapse HB/P/D into a single Immunisation (simple average of Hb, P, and D) and retain UD (dropping ID) alongside AM, HIV, and IC. The statistics are basically unchanged ($R^2 \approx 0.89, adjusted\ R^2 \approx 0.88$), but interpretation improves. After aggregation, Immunisation becomes significant and positive ($p \approx 0.0007$) with shared signals, while adult mortality, HIV, and income composition remain strong and theory-consistent. This demonstrates that adding overlapping variables harms precision and interpretability. The final model is therefore comprehensive but interpretable, isolating the levers to lift life expectancy: reduce adult mortality, lower HIV, lift immunization, and strengthen income composition.

MODEL EVALUATION AND DIAGNOSTICS

This section evaluates the final regression model developed in Part E, focusing on its statistical validity, diagnostic robustness, and real-world interpretability. We assess whether key regression assumptions are satisfied, analyse model performance metrics, and interpret coefficients within the context of global life expectancy drivers.

1. Regression Assumptions Diagnostics

Linearity:

Scatter plots of observed versus predicted life expectancy show a clear linear trend, indicating the model captures the central relationship between predictors and the dependent variable.

Independence of Errors:

Each observation represents a different country in 2015. However, cross-sectional data do not guarantee independent errors: countries may share regional conditions, economic relationships, or health shocks. Independence is therefore an assumption that has not been established by this analysis.

Homoscedasticity:

Residual plots exhibit some minor variance spreading as fitted values increase, but no severe systematic patterns. A potential remedy includes applying robust standard errors.

Normality of Residuals:

The Q-Q plot suggests that most residuals follow an approximately normal pattern, with some deviations in the tails. A large sample size does not make the residuals normally distributed; the Central Limit Theorem concerns sampling distributions rather than the distribution of individual residuals.

2. Model Performance Evaluation

The final regression model (Model B) shows a strong predictive capacity:

R-squared: 0.886
Adjusted R-squared: 0.882
F-statistic: $p < 0.001$ (highly significant)
Root Mean Square Error (RMSE): approx. 2.6 years

Model B explains approximately 88.6% of the observed variation in life expectancy among the 163 countries analysed. Its predictors are adult mortality, under-five deaths, HIV/AIDS, income

composition of resources, and the combined immunisation measure. This is an in-sample measure of fit, rather than evidence of predictive accuracy on new data.

3. Coefficient Interpretation

Income Composition (β = 33.2, p < 0.001):
A 0.1 increase in this variable corresponds to an approximate 3.3-year increase in life expectancy. This highlights the importance of equitable resource distribution.

Adult Mortality Rate (β = -0.0178, p < 0.001):
For every additional adult death per 1,000 population, life expectancy drops by 0.018 years.

HIV/AIDS (β = -0.688, p < 0.01):
Indicates the severe impact of infectious disease on lifespan, especially in less developed regions.

Under-five Deaths (β ≈ -0.002, p > 0.05):
The coefficient is negative but not significant after accounting for other mortality and health predictors, likely due to overlapping variance with adult mortality.

Immunisation (β ≈ 0.039, p < 0.01):
A single variable combining Hepatitis B, Polio, and DTP3 vaccination rates. A one percentage point rise in average immunisation is associated with a 0.04-year increase in life expectancy. The aggregation improves interpretability and statistical precision.

4. Conclusion

Model B provides a simpler summary of associations between life expectancy and the selected predictors, while retaining a similar in-sample fit to Model A. Adult mortality, HIV/AIDS, income composition, and immunisation are statistically significant predictors in this model. Diagnostic plots suggest some departures from the modelling assumptions, so results should be interpreted cautiously. The analysis does not establish causality, and predictive performance on new data has not been evaluated.

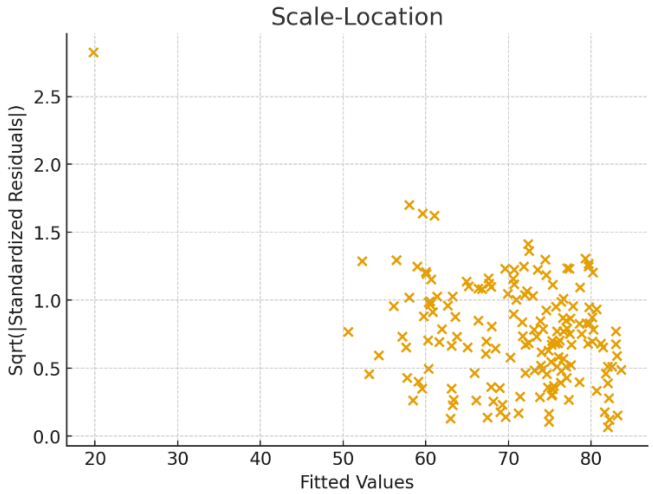


Figure 1: Residual Spread versus Fitted Values (Homoscedasticity Check)

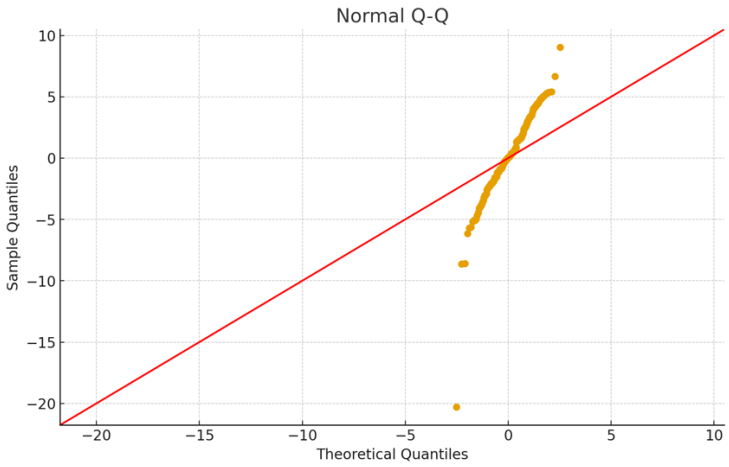


Figure 2: Normal Q-Q Plot (Assessing Normality of Residuals)

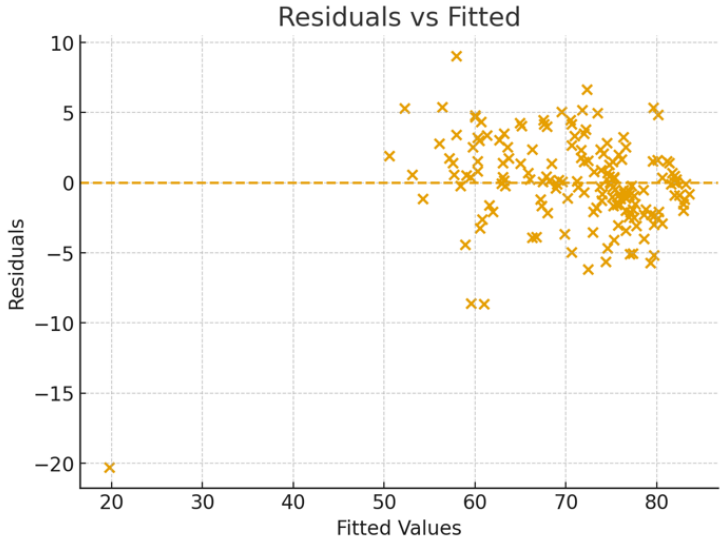


Figure 3: Residuals versus Fitted (Assessing Linearity and Model Fit)

POLICY ANALYSIS AND PREDICTION

South Africa offers a compelling case for policy analysis, as it exhibits relatively high national income and infrastructure with persistent health challenges and inequality. Despite its position as a middle-income developing country, its average life expectancy in 2015 was 62.9 years, which is significantly below the global average of approximately 71.3 years. This discrepancy provides an opportunity to explore how factors such as socioeconomic inequality and health challenges limit population longevity.

South Africa exhibits a complex life expectancy profile driven by health outcomes, disparities in income, and inequality in public services. Adult mortality remains high at 328 per 1,000 individuals, mirroring the impact of HIV/AIDS, which remains a dominant driver of premature death at a rate of 3.6% - one of the highest. Infant and under-five deaths were 42 and 52 per 1,000, respectively, which indicates a national improvement since the early 2000s, yet remain above global means for upper-middle-income countries, revealing potential barriers in child and maternal healthcare systems (World Health Organization, 2025). Immunisation coverage for Hepatitis B, Polio, and Diphtheria each stands at 75%, indicating substantial coverage, yet insufficient protection in relation to the 90-95% threshold required to achieve herd immunity (McDermott, 2021).

South Africa's average BMI in 2015 was 51.1, which is significantly higher than the global mean of approximately 42.5, suggesting an obesity epidemic that may contribute to rising rates of diabetes, hypertension, and cardiovascular disease - all relevant determinants of adult health and longevity (Centres for Disease Control and Prevention, 2022). Simultaneously, South Africa presents thinness rates of 4.4% for those aged 1-19, and 5.3% for those between 5-9 years old, displaying potential inequality in child nutrition. This data remains close to the global mean for thinness, which is 4.7% and 4.8% for these age groups, respectively. South Africa presents a score of 0.665 for income composition of resources, indicating moderate human development, slightly below the global mean of 0.684. This score aligns with the nation's well-evidenced condition of socioeconomic inequality, exhibiting a high Gini coefficient of 0.65 in 2015 - driven by ongoing ramifications of Apartheid, contributing to a significant racial wealth gap, unemployment, and income inequality (Statistics South Africa, 2020). South Africa's average schooling level is 13.0 years, which indicates broad access to secondary education and increasing tertiary participation. This figure aligns with the global mean of 12.8 years in 2015, indicating that education access is relatively strong and not among the most critical areas requiring improvement.

Therefore, South Africa's life expectancy dataset reveals a significant need to reduce adult mortality levels and expand comprehensive HIV/AIDs prevention and treatment coverage. Other primary improvement areas include immunisation coverage and public health infrastructure, to raise Hepatitis B, Polio, Diphtheria, and Measles immunisation rates to above 90%, and the implementation of nutrition programs to target both underweight children and obesity in adults.

Policy Recommendations and Predictive Analysis

Policy 1: HIV/AIDS Intervention

Model B ($R^2 = 0.886$ and Adjusted $R^2 = 0.882$) explains 89% of life expectancy variation, with HIV/AIDS emerging as a key driver. In South Africa, where adult prevalence is approximately 19.6% (World Bank, 2015), reducing this burden would yield substantial life expectancy gains. Model B's strong negative coefficient ($\beta = -0.688$; $p < 0.01$) indicates that each 1-unit reduction in the HIV/AIDS burden increases life expectancy by approximately 0.69 years.

1. **Expand HIV Testing and Early Diagnosis:** Early diagnosis is crucial for preserving immune function, allowing for a near-normal life expectancy, and for reducing the risk of transmission once treatment has been administered - particularly as many infections are transmitted by individuals unaware of their HIV status (May, 2017).
    - Implement universal opt-out HIV testing to all patients in public health facilities.
    - Introduce mobile testing units, broadening reach to include rural, mining and other non-urban communities.
    - Integrate HIV testing into sexually transmitted infection (STI), tuberculosis (TB) and pregnancy services to detect undiagnosed cases.

2. **Expand and Improve Existing ART (antiretroviral therapy) programs:** Implemented in 2004, the South African government has provided free ART through public healthcare facilities. By 2015, close to 3 million people were on ART, the highest number in the world. The program has resulted in a marked increase in life expectancy, from 56 to 61 between 2009-2012, and a 25% decrease in child and infant mortality rates in this same period (Simelela et al., 2015).
    - Adopt practices of treatment immediately after diagnosis, rather than waiting for disease progression.
    - Expand responsibility for treatment administration to include nurses in clinics (NIMART program), and pharmacists alongside doctors, improving access in rural areas and providing new avenues for care access.
    - Leverage digital tools such as artificial intelligence chatbots to distribute post-discharge messaging.

3. **Strengthen Prevention and Education Measures**
    - Education: implement education programs in secondary schools and tertiary institutions to reduce misinformation and stigma around HIV/AIDS, foster awareness and knowledge around symptoms, causes, prevention, testing, and healthy relationships.
    - Increase access to PrEP (pre-exposure prophylaxis): daily medication for HIV-negative people at high risk, including young women, sex workers, men who have sex with men (Healthdirect Australia, 2024).
    - Increase access to PEP (post-exposure prophylaxis): short-term ART taken within 72 hours after possible exposure - potentially needed in instances of sexual assault, needlestick injury (Healthdirect Australia, n.d.)

Feasibility and Cost-Effectiveness: This implementation strategy builds upon South Africa's already established ART program, utilising pre-existing infrastructure and systems, making this strategy feasible and less expensive than beginning an entirely new program. Further, the World Health Organisation (2014) found that ART and HIV prevention programs are highly cost-effective, typically below one times GDP per capita per DALY averted in sub-Saharan Africa.

Policy 2: Adult Mortality Reduction

In 2015, the primary causes of adult mortality in South Africa were communicable diseases and noncommunicable diseases, with specific leading causes including infectious diseases such as HIV/AIDs and tuberculosis, cardiovascular diseases and injuries from road traffic accidents and assaults. Adult mortality therefore interacts with variables such as HIV/AIDs and BMI, which are preventable causes. This policy recommendation will focus on injury-related deaths, from road traffic crashes, interpersonal violence and occupational incidents.

1. **Road Safety Measures:** The WHO Global Status Report on Road Safety in 2015 identified South Africa as having one of the highest road traffic fatality rates in the world, approximately 25 deaths per 100,000 population. This data exposes the need to strengthen road safety measures. This may be achieved through greater enforcement of speed limits, random breath testings, seatbelt and helmet laws to reduce road traffic fatalities.

2. **Strengthen Emergency and Trauma Systems:** Adult mortality rates may also be reduced by improvements in Emergency Medical Services (EMS), trauma care capacity and the development of reliable referral networks linking community, district and tertiary hospitals.
    - Strengthen EMS - increase ambulance coverage, particularly in rural areas, locate motorbike paramedics or community first-responder programs in dense urban traffic zones. Improved EMS coverage and rapid referral times has been found to reduce preventable trauma deaths by 15-30%, based on global benchmarks (Pham, Puckett, & Dissanaike, 2017). As trauma disproportionately affects adults aged 20-49, which is a key demographic in adult mortality statistics, reducing injury deaths directly lowers adult mortality rates which is a key variable in regression Model B ($\beta = -0.0178$).

Feasibility and Cost-Effectiveness: South Africa's existing EMS networks provide foundational infrastructure which can be upgraded as opposed to built from scratch, strengthening feasibility and improving cost-effectiveness.

Predictive Analysis

Model Choice Justification: Model A included eight predictors, including highly correlated variables such as infant deaths and under-five deaths. Model B reduced the number of predictors to five, with an R^2 of 0.8860 and an adjusted R^2 of 0.8823, compared with Model A's R^2 of 0.8898 and adjusted R^2 of 0.8841. Model B therefore provides a simpler specification with a small reduction in in-sample fit. These results describe cross-country associations; they do not establish the causal effects of health interventions or demonstrate predictive accuracy on new data.

HIV/AIDS

- The Model B coefficient ($\beta \approx -0.688$, $p < 0.01$) indicates that every one-unit decrease in the HIV/AIDS burden increases life expectancy by 0.7 years.
- Intervention scenarios:
    - Moderate reduction (–2 units) leads to an additional ≈1.4 year in life expectancy.
    - Ambitious reduction (–3 units) leads to an additional ≈ 2.1 years in life expectancy

Adult Mortality

- Coefficient **(**$\beta \approx -0.0178$, $p < 0.001$) shows that lowering adult mortality by 10 deaths per 1,000 predicts roughly 0.18 years of additional life expectancy
- By shortening emergency-response times and improving trauma coordination, South Africa could feasibly achieve a 1-2-year national life-expectancy gain within a medium-term horizon
- Because HIV/AIDS contributes heavily to adult mortality, improvements in HIV prevention and treatment also indirectly strengthen this pathway. This is an instance of multicollinearity between these two predictors

Limitations and Uncertainty

- Model B's residual standard error is approximately 2.62 years, describing the scale of residual variation in the fitted sample. It is not a prediction interval. A prediction interval for an individual country requires a separate calculation that accounts for both residual variation and uncertainty in the estimated mean response.
- Cross-sectional constraints - associations do not prove casualty, and improvements may appear gradually over several years
- Interdependence - HIV/AIDS and adult-mortality effects overlap, so combined impacts are not strictly additive

SUMMARY

The results of this report reveal that life expectancy is strongly influenced by the distribution of income-based resources, adult mortality rates, and the prevalence of HIV/AIDS. Through analysis of South African life expectancy outcomes, it is revealed how high adult mortality and infectious disease pressures constrain gains in life expectancy. Therefore, targeted public health interventions that expand treatment access, strengthen emergency healthcare responses and reduce preventable mortality are essential to improving life expectancy.

REFERENCES

Ahad, N. A., & Syed Yahaya, S. S. (2014). *Sensitivity analysis of Welch's t-test*. *AIP Conference Proceedings, 1605*, 888–893. https://doi.org/10.1063/1.4887707

Healthdirect Australia. (2024, September). *Pre-exposure prophylaxis (PrEP)*. https://www.healthdirect.gov.au/Pre-exposure-prophylaxis-PrEP

Healthdirect Australia. (n.d.). *PEP (post-exposure prophylaxis)*. Retrieved November 6, 2025, from https://www.healthdirect.gov.au/pep-post-exposure-prophylaxis

May, M. T. (2017). *Better to know: The importance of early HIV diagnosis. The Lancet Public Health, 2*(1), e6-e7. https://doi.org/10.1016/S2468-2667(16)30038-X

McKinsey Global Institute. (2023, June 29). *Pixels of progress: Chapter 3 – A dividend paid in years: Getting more health from each dollar of income.* McKinsey & Company. https://www.mckinsey.com/mgi/our-research/pixels-of-progress-chapter-3

Peters, D. H., Garg, A., Bloom, G., Walker, D. G., Brieger, W. R., & Hafizur Rahman, M. (2008). Poverty and Access to Health Care in Developing Countries. *Annals of the New York Academy of Sciences*, 1136(1), 161–171. https://doi.org/10.1196/annals.1425.011

Pham, H., Puckett, Y., & Dissanaike, S. (2017). Faster on-scene times associated with decreased mortality in helicopter emergency medical services (HEMS) transported trauma patients. *Trauma Surgery & Acute Care Open, 2*(1), e000122. https://doi.org/10.1136/tsaco-2017-000122

Preston, S. H. (2007). The changing relation between mortality and level of economic development. *International Journal of Epidemiology*, 36(3), 484–490. https://doi.org/10.1093/ije/dym075

Road Traffic Management Corporation. (2016, March). *Road traffic report, 1 October – 31 December 2015 (3rd quarter 2015/16)*. https://www.rtmc.co.za/images/rtmc/docs/traffic_reports/fqyr/3rd%20quarter%20201516%20Report.pdf

Simelela, N., Venter, W. D. F., Pillay, Y., & Barron, P. (2015). A political and social history of HIV in South Africa. *Current HIV/AIDS Reports, 12*(2), 256-261. https://doi.org/10.1007/s11904-015-0259-7

World Bank. (n.d.). *Prevalence of HIV, total (% of population ages 15–49) (Indicator SH.DYN.AIDS.ZS)*. World Development Indicators. https://data.worldbank.org/indicator/SH.DYN.AIDS.ZS?locations=ZA

World Health Organization. (2014). *Making fair choices on the path to universal health coverage: Final report of the WHO Consultative Group on Equity and Universal Health Coverage* (Publication No. 9789241507158). https://www.who.int/publications/i/item/9789241507158